

# Phase 1 - Week 3 - Lesson Day 4

Full EDA (Exploratory Data Analysis)

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## Today's Objectives

This is the final lesson day of Week 3, and it doesn't introduce brand new tools - it combines everything from this week (and Week 2's cleaning skills) into one structured workflow called **Exploratory Data Analysis (EDA)**. EDA is the real first step of almost every professional data science project: before modeling anything, you deeply understand the data first.

- Understand the standard structure of an EDA workflow
- Combine visualization and statistics into one coherent analysis
- Practice writing observations, not just producing charts

## 1. What Is EDA, and Why It Matters

**Definition:** *Exploratory Data Analysis* is the process of investigating a dataset using summary statistics and visualizations, before doing any serious modeling, in order to understand its structure, spot problems, and form early hypotheses about what the data is telling you.

**Why it matters:** skipping straight to modeling without EDA is one of the most common mistakes beginners make. You might miss a column full of outliers, a skewed distribution that needs transforming, or a strong relationship between two variables that should shape your whole approach. Every professional data scientist spends real time here before touching a model.

## 2. The Standard EDA Workflow

There's no single "correct" order, but a reliable structure to follow is below - this is essentially combining Week 2's first-look commands with Week 3's charts and statistics into one pass through the data.

1. Load the data and get a first look (`.head()`, `.info()`, `.shape`) - from Week 2, Day 1.
2. Check for missing values and duplicates, and clean if needed - from Week 2, Day 3.
3. Look at descriptive statistics for every numeric column (`.describe()`, plus mean/median/std individually where useful) - from Week 3, Day 3.
4. Visualize distributions of key numeric columns with histograms - from Week 3, Day 1.
5. Visualize relationships between numeric columns with a correlation heatmap and/or pairplot - from Week 3, Day 2.
6. Visualize categorical columns with bar charts, and compare numeric spread across categories with box plots - from Week 3, Day 1 and Day 2.
7. Write down your observations in plain language - what stands out, what looks skewed, what seems related to what, and what you'd want to investigate further.

### 3. Writing Observations (The Part Beginners Skip)

**Definition:** An observation is a short, plain-language statement connecting what a chart or statistic actually shows to what it means about the data. Producing a chart is not EDA by itself - the thinking you do about what the chart tells you is the actual value of the analysis.

Weak observation (just restates the chart):

```
"The histogram shows exam scores."
```

Strong observation (interprets it):

```
"Exam scores are roughly normally distributed around 70, with a standard deviation of about 10.  
A small cluster of students scored below 50, which might be worth investigating separately -  
possibly a subgroup that needs different support."
```

**Habit to build:** after every chart or statistic you produce from today onward, force yourself to write one sentence answering "so what?" - what does this actually tell me, and what would I want to check next?

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### Today's Exercise (Mini Challenge): Full EDA Report

Open **day-4-eda-project.ipynb**. This notebook walks through a complete, fully solved EDA on a generated sample dataset, following the exact 7-step workflow above - this is your template for running the same process on your own real dataset afterward.

After reviewing the solved walkthrough, repeat the same 7 steps on your own dataset from Week 2 (Day 1-3) in a copy of the notebook, writing your own observations at each step rather than reusing the example's.

#### Checklist (apply each step to your own dataset)

- Load the data and run `.head()`, `.info()`, `.shape`
- Check and handle missing values / duplicates
- Run `.describe()` and calculate mean/median/std for at least 2 key columns
- Plot histograms for at least 2 numeric columns
- Plot a correlation heatmap (and/or pairplot) for your numeric columns
- Plot at least one bar chart and one box plot involving a categorical column
- Write at least 5 plain-language observations across the whole analysis

Once you and Halimo have completed your own full EDA report, this wraps up Week 3 - next up is Week 4: Probability & Hypothesis Testing.